

Practical - 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as collapsible FAQ, modal popup,

image/content slider, notification banner, hamburger menu, and light/dark

theme switcher using JavaScript.

A. Theory

1. How are DOM elements selected and modified?

DOM elements are selected using JavaScript methods such as `getElementById()`, `querySelector()`, and `querySelectorAll()`. They are modified using properties and methods such as `textContent`, `innerHTML`, and `classList`. In our website, these methods are used to update the theme, display messages, and handle form validation.

Reference: W3C – Web Standards and DOM

2. Are event listeners attached correctly?

Yes. Event listeners are attached using `addEventListener()`. In our website, event listeners are used for actions such as the theme toggle and registration form submission. This allows the webpage to respond to user actions without putting JavaScript directly inside HTML elements.

Reference: W3C – Web Programming Guidelines

3. Is localStorage used for remembering the theme choice?

Yes. `localStorage` is used to save the user's selected theme. The website stores the theme as `studentHubTheme` and reads it when the page loads, so the selected light or dark mode remains after refreshing or reopening the page.

Reference: W3C – Web Storage and Client-Side Storage

4. Does interactivity improve usability without breaking accessibility?

Yes. Features such as theme switching and form validation improve usability by making the website easier to interact with. The website uses standard HTML controls such as buttons, inputs, labels, and forms, which helps maintain accessibility. W3C accessibility guidelines also recommend that interactive elements should be usable through different input methods, including the keyboard.

Reference: W3C – Web Content Accessibility Guidelines (WCAG)

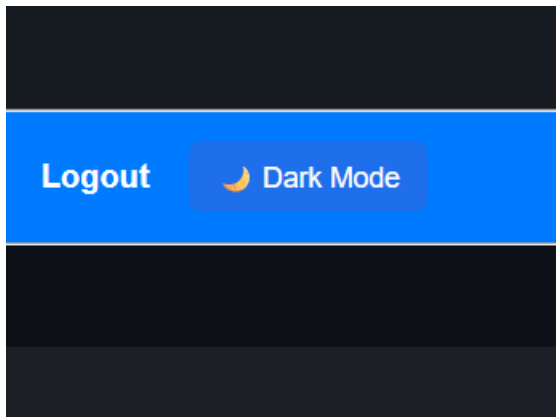
B. Implementation/ Output/Result

Result

The StudentHub pages were successfully changing its theme from dark mode to light mode perfectly without any concerns

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C. Evidence of Work (Screenshots)



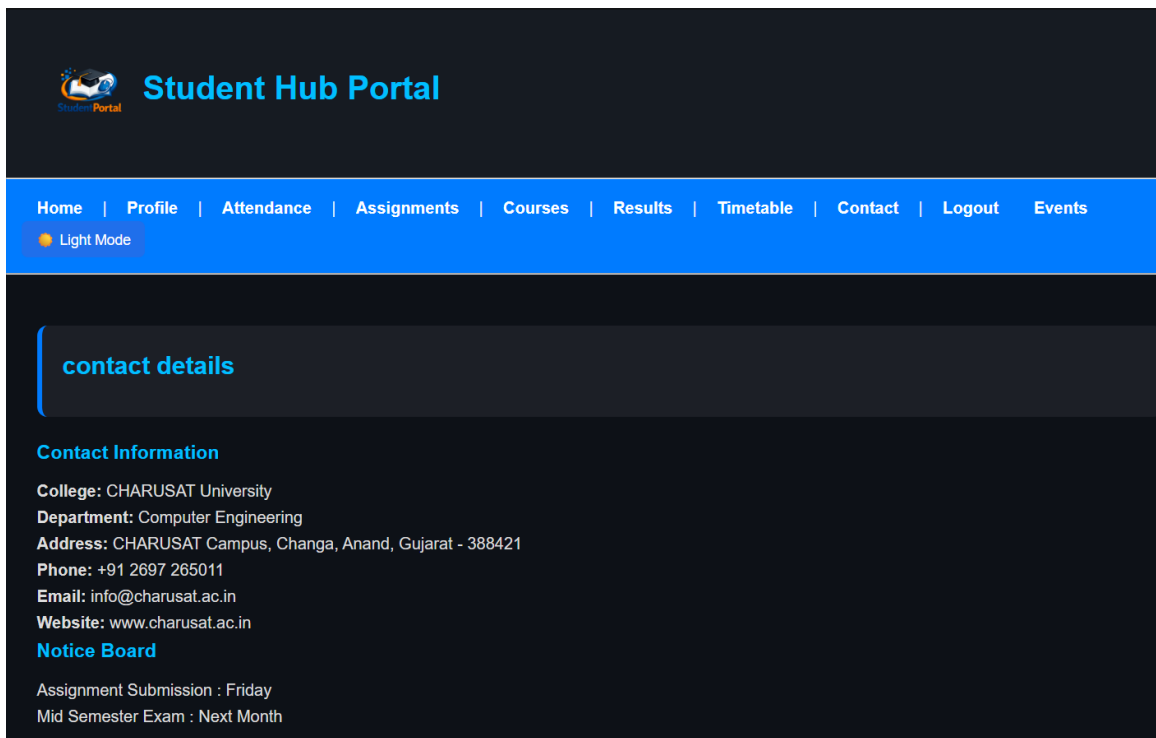
Student Hub Portal

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Dark Mode

contact details

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Assignment Submission : Friday
Mid Semester Exam : Next Month



The screenshot shows the StudentHub dashboard in both dark and light mode.

D. GitHub Repository Link:

GitHub Repository: <https://github.com/rehansaiyed72/student-hub-portal>

E. Challenges Faced:

When I made the dark and light mode the code and all were perfect but still button was not working at all as if it was unclicable then I added pop up notification with button as button is clicked then it started working

F. Learning Outcomes:

After this practical I learn how to use java script toggle the screen add drop down menu and etc.